Disparity Analyses Are Robust To Ancestral State Estimation Uncertainty

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Abstract

Ancestral state estimation is a widely used method to infer unobserved traits across a phylogeny, extending analyses beyond incomplete fossil records. In studies of morphological disparity ancestral state estimation can account for ghost lineages, yet estimates are often confounded with high uncertainty. How best to deal with this uncertainty remains unclear. Using simulations, we compared pre- and post-ordination estimation methods, as well as both point estimate and probabilistic approaches for treating uncertainty, under varying transition rates and fossil sampling levels. The relative disparity error was calculated from the difference between our estimated disparity values and the true disparity values from the original simulations. We then extended this framework to evaluate the recovery of changes in disparity across time by simulating mass extinction events with varying intensity and selectivity. We find that pre-ordination probabilistic estimations outperform post-ordination and point estimation methods and that omission of ancestral state estimations leads to underestimates of disparity. As expected, higher fossil sampling improves estimation accuracy, although recovered disparity accuracy was robust even at low fossil sampling. Similar patterns were observed when recovering changes in disparity across mass extinction events but declined significantly when fossil sampling was low, highlighting that fossil inclusion is essential when assessing disparity in deep time. These results demonstrate that explicitly propagating uncertainty, rather than collapsing states into point estimates, is more likely to recover accurate patterns in disparity, reinforcing the need to move beyond point estimates when working with the incomplete fossil record.

Keywords: ancestral state estimation; morphological disparity; fossils; simulation; uncertainty; macroevolution

1 Introduction

2 Materials and Methods

2.1 Simulation Framework

2.1.1 Phylogenies

Simulations were carried out using treats (Guillerme 2024). Phylogenies were simulated using a modified version of the birth-death process algorithm. The process begins with a single lineage and iteratively selects lineages to grow the tree forward in time. At each step, all living branches are extended by a branch length increment, drawn from an exponential distribution based on the total number of lineages and the combined speciation and extinction rates. Selected lineages then undergo either speciation or extinction based on a probabilistic decision determined by the specified rates. Phylogenies were simulated with birth-death parameters as speciation (λ) = 1.0, and extinction (μ) across a range between 0.9-0.5. Fixing $\lambda = 1.0$ provides a convenient relative time-scale, since only the ratio of extinction to speciation affects the shape and balance of simulated trees. The chosen parameters correspond to a moderate-to-high turnover regime, consistent with the upper-empirical estimates from both fossil- and molecular-based diversification studies of vertebrate groups. This level of extinction produces realistic evolutionary trees with substantial lineage pruning and extinct diversity, while maintaining a positive net diversification rate (λ - μ = 0.3). However, these parameters were not fixed, since they can vary if a tree fails to be generated with given stop rule and parameters. A histogram of the distribution of speciation/extinction ratios is in the supplementary. Trees were simulated at three different sizes: 50, 100 and 150 living taxa.

2.1.2 Trait simulation

Across each of the phylogenies, we simulated discrete and continuous trait data separately. Discrete trait data were simulated across three rate categories, representing fast, medium and slow transition rates. These rates were assigned by generating equal rate transition probability matrices 1. For each category, 85 binary characters and 15 three-state characters were simulated, representing the coding scheme found in most discrete datasets. Discrete trait evolution was simulated using a discrete-time Markov process, where the probability of state transition along each branch was proportional to the product of the transition probability and branch length.

Continuous trait data were simulated under variations of Brownian motion and Ornstein-Uhlenbeck models. For Brownian motion, the trait data was simulated as both default and with a moving trend. For Ornstein Uhlenbeck, the data was simulated as strong, weak and with differing optimum and root states. Following Gearty et al. (2024), the alpha (α) parameter was set based on phylogenetic half life ($ln(2)/\alpha$) to account for differing tree heights across simulations (Cooper et al. 2016). Continuous trait data were simulated across 100 characters for each model across each phylogeny.

2.2 Fossil sampling

In order to simulate different levels of fossil sampling, each tree was split into four chronological 'stages' to replicate the stratigraphic nature of the fossil record. Fossil tips were then subsampled so that only a percentage of fossils are present, ω (where $\omega_P = P\%$ fossils sampled). Five different fossil sampling levels were generated for each simulated phylogeny, ω_0, ω_5, ω_{15}, ω_{50} and ω_{100}. The simulated trait matrices were analogously downsampled and node states removed.

2.3 Ancestral state estimation

Ancestral state estimation was carried out using Maximum Likelihood. We used four combinations of two approaches to ancestral state estimation to understand which was the best at recovering the true disparity signal. Firstly, we compared pre- versus post-ordination estimation methods. Pre-ordination estimation methods of discrete characters involves estimating the marginal likelihoods for each discrete state at each node, based on a Maximum-Likelihood derived estimate of the Mk transition matrix and the tree topology given (Pagel 1999; Yang 2006).

2.3.1 Discrete estimation

2.4 Disparity Calculation

Morphospaces were constructed using principal coordinates analysis (PCO). Disparity was quantified using two metrics: sum of variances and mean nearest-neighbor distances. Relative disparity error was calculated as:

$$\text{Relative Error} = \frac{\text{Estimated Disparity} - \text{True Disparity}}{\text{True Disparity}} \tag{1}$$

2.5 Statistical Analysis

3 Results

3.1 Overall Performance of ASE Methods

3.2 Sensitivity to Evolutionary Rate

4 Discussion

5 Conclusions

Pre-ordination probabilistic ancestral state estimation provides the most accurate approach for morphological disparity analysis. Explicitly accounting for uncertainty in ancestral reconstructions is crucial for robust disparity estimation, particularly when fossil sampling is limited.

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Author Contributions

C.S. conceived the study, performed analyses, and wrote the manuscript. N.C., G.T., and T.G. contributed to study design and manuscript revision. All authors approved the final version.

Data Availability

Data and code are available at: [repository URL]

Competing Interests

The authors declare no competing interests.

Figures

Tables

Table 1: Trait simulation models and parameters used across each phylogeny. For continuous characters, the root state was always set as 0.

Type of data	**Model**	**Parameters**
Discrete	Fast	$P_{ij} = 0.9$
	Medium	$P_{ij} = 0.1$
	Slow	$P_{ij} = 0.01$
Continuous	Brownian motion	$\sigma = 0.25$
	Brownian motion (trend)	$\sigma = 0.25,\ \mu = 0.1$
	OU (strong)	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height / 10}}$
	OU (weak)	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height / 5}}$
	OU (shift optimum)	$\sigma = 0.25,\ \alpha = \frac{\ln(2)}{\text{tree height / 5}},\ \theta = 2$

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